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AI-Assisted Reconstruction of Past Environments

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Abstract. Reconstructing paleoenvironmental conditions in archaeology is key to understanding past human-environment interactions. However, this process often involves extensive, repetitive, and time-consuming data analysis. The introduction of Artificial Intelligence (AI) techniques can be a valid tool for assisting experts in managing this workload.
This project focuses on environmental archaeology, employing AI techniques (e.g., tree-based algorithms, neural networks) to predict biomes by analysing pollen counts collected during sampling or archaeological excavations.
Pollen counting involves quantifying the species or families in a sample, reflecting the vegetation composition and distribution during pollen deposition. Associating species to infer environmental conditions requires expert input and the consideration of multiple variables (e.g., soil characteristics and pollen dispersal) to explore human-environment dynamics over time.
The study integrates predictive and generative AI models to enhance data analysis. Predictive AI supports experts by accelerating species identification and association, while generative AI produces visually impactful reconstructions. The tool is developed collaboratively with professionals from diverse fields (e.g., archaeologists, mathematicians, botanists) to ensure scientifically valid results. Additionally, it serves as both an educational resource and a means for effective dissemination.

Keywords. Paleoenvironmental Reconstruction, Artificial Intelligence in Archaeology, Environmental Archaeology, Human-in-the-Loop

1. Research context

In recent years, integrating Artificial Intelligence (AI) into archaeology has opened new avenues for research and practice [1]. AI methods have shown promise in automating tasks such as cataloguing finds [2], analysing use traces on osteological materials [3], or studying the geological provenance of the lithics [4]. Similarly, AI-driven models aid site detection by evaluating ground markers [5][6].

However, the adoption of AI in archaeology introduces technical and ethical challenges [7]. Technically, the discipline faces issues such as limited access to primary data, inadequate data sharing [8], and the lack of standardised methodologies [9]. These limitations impede large-scale analyses and cross-comparative studies, particularly in a field characterised by the destructive nature of its methods. To address this from a data perspective, archaeologists have begun adhering to FAIR principles—findable,

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accessible, interoperable, and reusable [10][11]—and leveraging persistent digital infrastructures [12]. Ethically, it is central to maintain transparency in the reconstruction process [13]; AI-driven analysis results are not objective truths but interpretations shaped by the archaeologist's expertise, methodological choices, and the selected algorithms. One key to achieving transparency is the close collaboration between experts from different domains, in this case, mainly between archaeologists and computer scientists. While archaeologists provide domain expertise to contextualise and validate AI-generated results, computer scientists analyse the nature of the data to determine the most suitable algorithms.

Building on the foundational work of Sobol and Finkelstein [14] and Sobol et al. [15] in machine learning applications for biome reconstruction, this research aims to reconstruct past environments based on archaeological pollen data using two approaches. The first is a predictive model for biome classification in Europe, while the second, representing this study's innovative aspect, involves generative AI to create landscape images. This visual output can serve as an educational tool and, if sufficiently reliable, may also contribute to research. Open data repositories like Neotoma [16], SEAD [17], EMPD [18][19], and BRAIN [20] are used to build the datasets, while software tools such as QGIS, OpenRefine, and Google Colab facilitate data processing.

1.1. Algorithms and environmental reconstructions with pollen data.

Artificial Intelligence (AI) application in palynology has introduced innovative methods for reconstructing vegetation landscapes, complementing traditional approaches like biomisation [21]. Biomisation simplifies the complexity of biomes by linking pollen taxa to plant functional types and biome categories. While effective, this method relies on generalisation, which may limit its accuracy in capturing detailed ecological interactions.

Recent advancements have focused on algorithmic models developed within *Geographic Information Systems* (GIS) environments [22]. Sugita's *Landscape Reconstruction Algorithm* (LRA) was an essential contribution to the field, which integrates the *Regional Estimates of VEgetation Abundance from Large Sites* (REVEALS) model for regional vegetation estimates with the *LOcal Vegetation Estimates* (LOVE) model for local vegetation assessments. This dual-scale approach allows for detailed reconstructions of vegetation patterns at both macro and micro levels, and it has since become a standard in landscape reconstruction studies [23][24].

The increasing use of machine learning and AI has introduced new possibilities in palynology, mainly through neural networks and classification algorithms. Bourel et al. [25] demonstrated that using neural networks to automate pollen identification from microscopic images significantly reduces manual labour and improves efficiency.

In a landmark study, Sobol and Finkelstein [14] employed machine learning models, including Random Forest, to classify modern pollen samples from Africa and Arabia into biome categories, the study achieved high accuracy (F1 score: 0.85). It used 1125 samples categorised into nine biome types, later consolidated into five for statistical robustness. These efforts underscored the utility of non-parametric models like Random Forest, which outperform parametric models (e.g., logistic regression) in handling complex ecological datasets.

Building on this work, Sobol et al. [15] applied these machine-learning techniques to fossil pollen data from the Wonderkrater site in South Africa, covering a timeline of 60,000 years. By harmonising modern and fossil datasets and using Random Forest models trained on modern pollen records, the study successfully reconstructed vegetation changes from the Late Pleistocene to the Holocene. The model revealed shifts in biomes corresponding to climatic transitions, showcasing the potential of AI for long-term palaeoenvironmental analysis.

A similar study by Wang et al. [26] on the Tibetan Plateau, confirmed the potential of AI models in paleoenvironmental reconstruction. Using Random Forest algorithms, the researchers reconstructed biome changes over 22,000 years, correlating their findings with global climate shifts and variations in the Asian monsoon. These studies highlight the robustness and adaptability of AI-driven methods in addressing the complexities of palaeoenvironmental reconstruction.

In the observed studies, the importance of quality empirical data, modern and fossil, must be emphasised as a fundamental starting point, significantly influencing the accuracy of results.

2. Methodology

2.1. Dataset creation and preparation

The methodological framework for this study began with creating a comprehensive dataset that integrates data from leading palynological databases such as Neotoma, SEAD, and EMPD. The dataset encompasses modern and ancient pollen data and geographic metadata like latitude, longitude, and site identifiers.

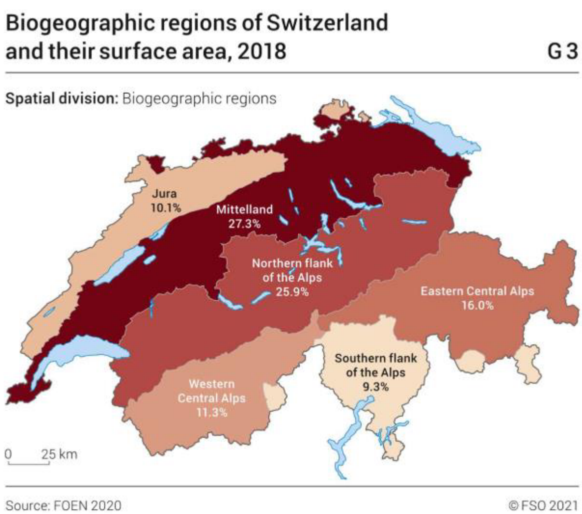

Figure 1. Biogeographic regions of Switzerland. Source: FOEN 2020

The first phase of the study focused on Switzerland as a test case, where detailed modern pollen data were analysed. Switzerland was chosen because it represents a relatively well-defined area, allowing for a structured approach to data processing. Additionally, it is well-documented in terms of available sources and exhibits a good diversity of biomes, making it a suitable starting point for methodological development. The Swiss dataset includes 75 modern sites, 207 ancient sites (dating back to 38,600 years), and 12 sites with entomological records spanning 2,500 to 13,500 years. An identification sheet was compiled for each pollen species, documenting key attributes such as habitat, biome, pollen production, and other relevant characteristics. The region was divided into six subregions (Figure 1), and the initial analyses were conducted exclusively on modern pollen data (i.e., dating to the last 500 years) to establish a baseline for further research. Tree-based algorithms, including decision tree classifiers, were applied to this dataset, demonstrating initial success in bioregions classification. These tests provided valuable insights into model performance on small, region-specific datasets.

Building on the insights gained, we are working to expand the dataset to include sites across Europe due to the need for more data for the AI algorithms. So far, 35,000 analysable samples have been identified from 3,100 sites, containing over 6,000 species. This broader dataset increases biome diversity, incorporating records from regions with distinct ecological and climatic histories. Efforts are underway to standardise the dataset to meet algorithmic requirements and address specific archaeological research questions. Key ecological references will be used to guide standardisation [27][28][29], ensuring consistency in biome classification across diverse regions.

As the dataset expands, the methodological approach emphasises the integration of human expertise; Archaeologists, archaeobotanists, and botanists actively refine data normalisation processes, ensuring the dataset meets computational needs and archaeological objectives. For instance, altitude, temperature, and human activity indicators are incorporated to enhance contextual accuracy. Specifically, after discussions with archaeologists and paleobotanists regarding selecting the area from which to build the dataset, the spatial range was expanded following the suggestion of computer scientists to reach a compromise on the amount of data to be acquired. Once the dataset was populated, all tree and plant species were reviewed with the support of a botanist, who provided descriptions and indicated each species' area of origin, period of introduction at the site, and associations with other species. The examination of species is essential for understanding the dataset's composition. These refinements are critical for distinguishing between naturally occurring pollen and that influenced by human activity, providing deeper insights into past human–environment interactions.

2.2. Predictive modelling

Table 1. Performance metrics of different machine learning models for the prediction of Swiss bioregions.

	Precision	**Recall**	**F1-score**	**Accuracy**	**95% Confidence Interval (Accuracy)**
Decision tree	0.81	0.76	0.78	0.79	[0.65, 0.93]
Random forest	0.89	0.74	0.77	0.81	[0.68, 0.94]
Gradient boosted trees	**0.91**	**0.80**	**0.84**	**0.91**	**[0.81, 1.00]**
SVM	0.70	0.42	0.49	0.53	[0.36, 0.70]
KNN	0.45	0.40	0.42	0.43	[0.26, 0.60]
Neural Network	0.65	0.58	0.60	0.75	[0.60, 0.90]

Table 1 presents the preliminary results of the classification task on the Swiss dataset, evaluating the performance of various machine learning algorithms in predicting bioregions based on pollen data. The dataset consists of 163 instances, each corresponding to a site, and was created using 421 features representing the relative abundance of plant taxa. After the initial expert-based refinement, conducted with the help of a botanist, the feature set was reduced to the 100 most informative variables, selected for their ecological relevance to the bioregions. Dimensionality reduction algorithms (e.g., Principal Component Analysis (PCA) or feature selection based on mutual information) will be tested on the updated dataset that is currently under development.

The target variable includes six Swiss bioregions: Jura (31 instances), Mittelland (6), Northern Flank of the Alps (Alpi Nord, 45), Eastern Central Alps (Alpi Est, 3), Western Central Alps (Alpi Ovest, 57), and Southern Flank of the Alps (Alpi Sud, 21). Although the classes Mittelland and Alpi Est are underrepresented, they were retained to ensure full geographical coverage and to assess model robustness in less common regions. The dataset was divided into 80% for training and 20% for testing, using stratified sampling to preserve the original class proportions. Model training and parameter tuning were conducted using 5-fold stratified cross-validation, applied solely to the training set.

The results indicate that tree-based algorithms outperform other approaches. Gradient Boosted Trees achieved the highest performance, with an accuracy of 91%, F1-score of 0.84, precision of 0.91, and recall of 0.80, along with a 95% confidence interval for accuracy of [0.81, 1.00]. Random Forest also performed well with 81% accuracy, while Decision Trees reached 79% with competitive F1-scores. In contrast, non-tree-based models such as Support Vector Machine (SVM), k-nearest neighbours (KNN), and Neural Networks exhibited substantially lower performance, with accuracy scores of 53%, 43%, and 75%, respectively. These results suggest that SVM and KNN struggle to model the complex, high-dimensional relationships between plant species distributions and bioregions, likely due to their limited capacity to handle ecological heterogeneity in sparse datasets. This preliminary test on modern pollen data provides a foundation for expanding the study to ancient pollen and a broader European dataset. However, while these analyses offer valuable insights, certain limitations must be considered for the project's future development. Dataset sampling biases, pollen production and dispersal variations, and the potential overrepresentation of certain bioregions may influence classification accuracy. Additionally, while modern pollen records provide a reliable basis for predictive modelling, extending this approach to ancient pollen introduces further challenges related to preservation biases, differential representation of taxa, and changes in deposition patterns over time. Despite these constraints, the results demonstrate the potential of these methodological approaches, reinforcing their applicability in future steps of this project.

To enhance the interpretability of our results, the predictions will be visualised through spatial mapping, facilitating a clearer understanding of regional biome shifts. Moreover, it highlights key aspects that must be carefully considered in the following steps, particularly the necessity of defining clear chronological frameworks that emphasise significant vegetation changes over time. These include the glacial period, the post-glacial period, the transition to human sedentism, and modern revolutions such as the discovery of the Americas and the Industrial Revolution. These temporal distinctions are crucial for accurately contextualising vegetation dynamics and human-environment interactions. This further underscores the importance of data normalisation, ensuring that pollen records from different periods remain comparable and that predictive models are both robust and meaningful.

Furthermore, the predictive outcomes will provide valuable input for the generative phase, aiding in the reconstruction of past landscapes. However, it is essential to acknowledge that these predictions are model-derived and, therefore, require careful validation. Their reliability will be assessed by comparing them with entomological data and through expert evaluation by a palaeontologist, whose insights will help refine the model's interpretations in the context of historical ecological conditions.

2.3. Generative modelling

Figure 2. Landscape reconstruction, different models, same inputs. Left: Image generated by the DeepAI Generator in October 2023. Right: Image generated by the Image Creator from Microsoft Designer (Bing) in October 2023.

Over the past year (2024), generative AI has been experimented with in archaeology to create illustrative images of scenarios, particularly in prehistory [30].

This study explored the capabilities of various AI models, such as ChatGPT 4.0 with DALL-E integration, DeepAI, Bing Image Creator, and MidJourney, to generate images based on botanical descriptions and pollen diagrams, highlighting the potential and the limitations of current AI tools in producing scientifically meaningful visuals.

At this stage, the use of generative AI remains preliminary, and a dedicated methodology still needs to be developed to ensure that the outputs are scientifically grounded and valid for use in archaeological and palaeoecological research. The idea is to create a link between this study's predictive and generative modelling components: we plan to use the outputs of the bioregion classification model (e.g., predicted biome label, site coordinates) as structured prompts for generative image creation.

As a first example, we tested AI-generated depictions of the vegetation of the Gulf of Baratti, located in the municipality of Piombino, Livorno. The input remained consistent and included the Latin names of the species present in the area: "The vegetation is characterised by the rocky coast, which is colonised up to a few metres from the sea by Foeniculum vulgare, Daucus carota, Helichrysum italicum and the silvery shrubs of Sempervivum tectorum, along with sporadic specimens of Pinus pinaster." This description is derived from a summary of the environmental characterisation provided on the official website of the Val di Cornia Parks (source, last accessed 30/01/2025). However, the generated images varied significantly in style and accuracy, often omitting particular species or introducing artistic elements not grounded in the provided data. Being a reconstruction of an environment that is no longer present, the evaluation of it is based on the user's knowledge, which leads to the need to pay special attention to the methods by which it is used.

In Figure 2, two AI-generated images from different models illustrate this variation. On the left, a photorealistic rendering depicts a meadow in the foreground with wild fennel and carrot plants, while the background features the sea and what appears to be a pine forest. On the right, the representation is more stylised and cartoonish, with abundant vegetation and bright colours. The fennel and pines are recognisable, and the shrubs exhibit a shape typical of coastal dunes. While these visualisations remain far from accurately depicting the plant species, they offer an initial approximation of the intended landscape.

Further reconstructions were conducted using ChatGPT 4, based solely on pollen diagrams containing quantitative records of pollen taxa found at specific sites. The initial AI-generated images tended to incorporate artistic liberties, such as introducing elements absent from the pollen data—mushrooms and incorrectly proportioned trees, for instance. However, these images became progressively more realistic and botanically accurate through an iterative dialogue with the AI. By engaging with the model (e.g., asking, "On what basis was the water included in the image?" or "Why were these particular colours chosen?"), experts were able to refine the outputs to better align with ecological data. This iterative process was guided by the involvement of archaeobotanists and archaeologists, who played a key role in both formulating the input (e.g., species selection, landscape setting) and critically evaluating the output, checking for botanical plausibility (e.g., species co-occurrence, visual appearance, correct scaling of plant sizes).

Beyond methodological considerations, these tools hold significant potential for education and dissemination. By visualising complex ecological data, AI-generated images can make palaeoecological reconstructions more accessible to broader audiences. Such images are handy in educational contexts, where they help bridge the gap between abstract scientific concepts and tangible representations. Moreover, as these models continue to improve in realism and accuracy, their application could extend to scientific research, offering novel ways to visualise and interpret ecological data. For example, AI-generated images might reveal previously overlooked patterns or relationships within datasets, providing fresh perspectives for researchers.

However, their effectiveness depends on maintaining a strong interaction between automated systems and expert knowledge, ensuring the generated representations remain ecologically meaningful and archaeologically relevant.

3. Conclusion

Reconstructing past environments requires evaluating multiple interacting variables that have evolved. Key factors include sampling location, climate variability, changes in wind patterns, and the impact of human activities. These parameters must be assessed within their temporal and ecological contexts, necessitating collaboration among specialists from different fields, including archaeobotanists, entomologists, and archaeologists, to ensure a scientifically robust reconstruction.

Data collection for this study involved consolidating information from diverse databases, such as Neotoma, SEAD, and EMPD. Each database presents unique structures and data coverage, requiring manual reconciliation to account for differences in information availability. This process has proven to be the most complex due to variations in data storage formats; however, it is essential for building a dataset suitable for predictive modelling.

The initial algorithm development phase focused on transparent models, beginning with Decision Trees and progressing towards Random Forest and gradient-boosted trees. These models were chosen for their interpretability, allowing a direct understanding of the classification processes and feature importance. They appear to be the most effective compared to more complex techniques tested (e.g., Neural Networks). On the other hand, generative approaches, such as GANs and Large Language Models (LLMs) like GPT-3, have demonstrated value in image generation but also present significant challenges related to explainability. The opacity of these models, compounded by proprietary training datasets, limits the ability to fully assess potential biases or data inconsistencies.

Future efforts will prioritise the development of custom models trained on open datasets to address these limitations. This approach will provide greater transparency and control over the training process, enabling researchers to tailor algorithms to specific ecological and archaeological questions. For example, integrating Stable Diffusion and leveraging APIs from advanced LLMs may offer a starting point for creating interpretable and scientifically rigorous models, particularly for generating images.

The interaction between text and image generation models has already demonstrated potential in reconstructing vegetation landscapes. The ability to interact with models in real time, providing iterative feedback and refining outputs, highlights the crucial role of human expertise in ensuring accuracy and contextual relevance. This collaborative approach allows experts to guide AI tools in producing visuals that closely align with empirical data while addressing inaccuracies and artistic liberties inherent in generative processes.

From a broader perspective, AI-driven tools offer valuable opportunities for education, dissemination, and scientific exploration. By visualising complex datasets, these technologies make palaeoecological reconstructions more accessible to diverse audiences, bridging the gap between abstract scientific concepts and tangible representations. As these models evolve, their potential applications in scientific research extend beyond visualisation to generate novel hypotheses and uncover previously unrecognised patterns within datasets.

Ultimately, AI's success in reconstructing past environments depends on maintaining a symbiotic relationship between human expertise and machine learning. This collaboration ensures that outputs remain scientifically credible and contextually meaningful, paving the way for new insights and methodologies in archaeology and palaeoecology.

This study represents an ongoing research effort with a methodology that continues to adapt based on the analysis of results, expert feedback, and the needs of specialists from different domains. We are currently working on normalising the chronological calibration to establish a dataset with comparable and usable samples for algorithm integration, thus proceeding with the predictive and generative phases of the project.

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